Code vs Text Analysis: Exploratory Data Analysis Using Python

- Exploratory Data Analysis (EDA)
- Dataset: StackOverflow + Twitch
- Goal: Identify differences between programming code and natural language text

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Dataset Overview

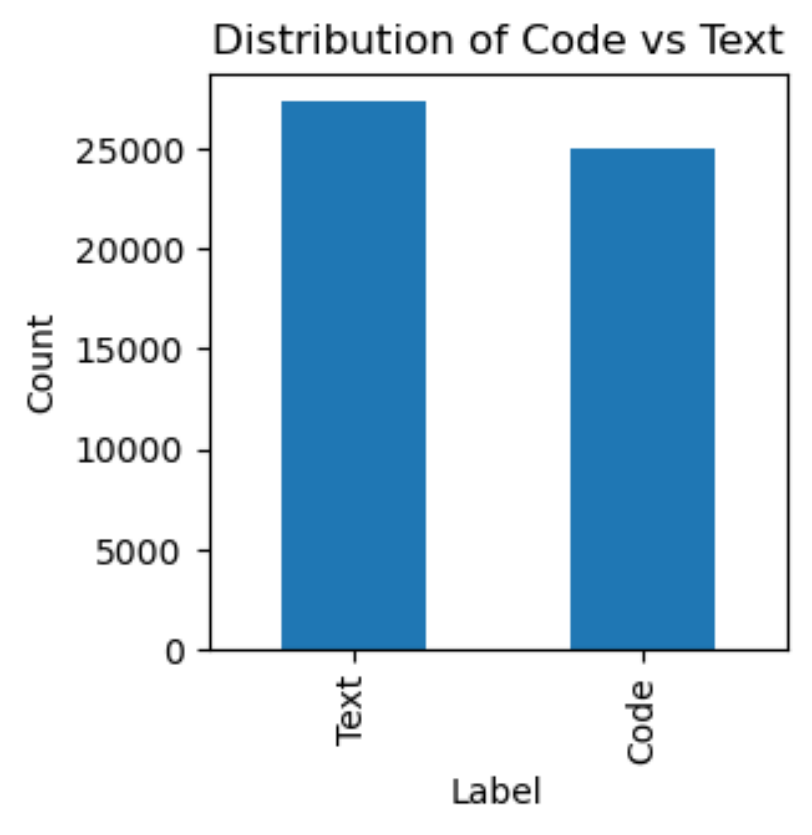
- Dataset: Code vs Text
- Rows: ~52k
- Labels: Code (1) and Text (0)
- Sources: StackOverflow and Twitch
- Task: Binary classification

```
In [3]: data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 52270 entries, 0 to 52269
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   line    52266 non-null  object
 1   label   52270 non-null  int64
 2   source  52270 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.2+ MB
```

```
In [6]: #Deleting rows foe all the 4 instances of null values in text column to clean the data.
        data = data.dropna()
        data.info()

<class 'pandas.core.frame.DataFrame'>
Index: 52266 entries, 0 to 52269
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   line    52266 non-null  object
 1   label   52266 non-null  int64
 2   source  52266 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.6+ MB
```

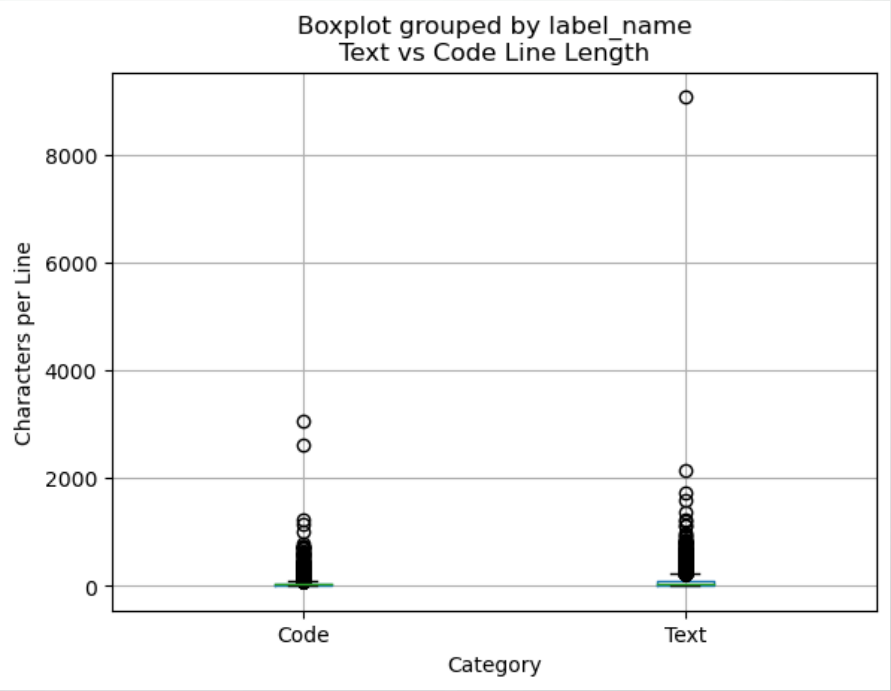
Label Distribution

- Text ≈ 27k lines
- Code ≈ 25k lines
- Dataset is fairly balanced
- Balanced datasets help ML models perform better



Characters in Text vs code

- Typically line contains more character than code
- Some extremely long lines exist, likely due to long code expressions or detailed textual explanations
- Code line lengths are more tightly distributed, while text line lengths vary significantly



Line Length Analysis

- Text mean length ≈ 78 characters
- Code mean length ≈ 39 characters
- The median line length for text is higher than code.
- Some extremely long lines exist, likely due to long code expressions or detailed textual explanations
- Code line lengths are more tightly distributed, while text line lengths vary significantly.

```
In [10]: data['line_length'].describe()

Out[10]: count    52266.000000
         mean        59.531741
         std         85.952399
         min          1.000000
         25%         17.000000
         50%         34.000000
         75%         72.000000
         max       9082.000000
         Name: line_length, dtype: float64
```

```
In [11]: data.groupby("label_name")["line_length"].describe()
```

Out[11]:

label_name	count	mean	std	min	25%	50%	75%	max
Code	24938.0	38.975660	47.750157	3.0	16.0	29.0	50.0	3062.0
Text	27328.0	78.290069	106.356024	1.0	18.0	46.0	107.0	9082.0

Top Code Tokens

```
In [15]: data.groupby("label_name")["symbol_count"].describe()
```

Out[15]:

label_name	count	mean	std	min	25%	50%	75%	max
Code	24938.0	3.120178	15.316965	0.0	1.0	2.0	4.0	2108.0
Text	27328.0	0.587602	2.445844	0.0	0.0	0.0	0.0	159.0

- Code contains more symbols
- Examples: {}, (), ;, =, <, >
- Text rarely contains programming symbols
- Symbol count helps distinguish code from text

Top words in text vs code (NLP insight)

Code lines are dominated by programming language syntax and keywords such as return, int, class, and public, which are essential components of programming languages.
Code lines frequently contain structural elements such as imports, variable declarations, and class definitions.

Text lines mainly consist of problem descriptions, debugging discussions, and explanations typical of programming forums.

Some technical terms appear in both categories because they are used both in code implementation and in textual explanations.

Out[29]:

	Top code words	count
0	java	961
1	return	906
2	int	816
3	org	715
4	class	662
5	public	640
6	const	561
7	app	520
8	data	516
9	string	492
10	id	491
11	new	467
12	import	464
13	div	459
14	internal	433
15	react	429
16	true	428
17	user	425
18	var	421
19	std	416

Out[31]:

	Top text words	count
0	code	1994
1	like	1566
2	error	1386
3	using	1372
4	following	1119
5	app	1052
6	use	955
7	way	938
8	tried	904
9	problem	878
10	want	806
11	file	799
12	just	785
13	function	734
14	data	720
15	does	716
16	example	699
17	time	674
18	trying	657
19	ve	657

Potential ML features

- Line length
- Symbol count
- Word count
- TF-IDF features
- These can be used to build a classification model

Insights

- Code lines are shorter but symbol-heavy
- Text lines are longer and language-based
- Programming keywords dominate code
- Discussion/problem words dominate text
- The difference in line length distribution between code and text suggests that line length could be an effective feature for building a classification model that distinguishes programming code from natural language.

Conclusion

- Code and natural language text show clear structural differences.
- Simple statistical and NLP features can effectively distinguish them.